Yuheng Wang

Ph.D. Student in Applied Physics
Rice University
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Education

- **Rice University** — Aug. 2026 – Present
  Ph.D. in Applied Physics — Houston, USA
- **University of Michigan** — Sep. 2024 - May. 2026
  Undergraduate in Engineering Physics, minor in Computer Science — Ann Arbor, USA
  GPA: 3.96/4.00; Summa Cum Laude
- **Shanghai Jiao Tong University** — Sep. 2022 - August 2026
  Undergraduate in Electrical and Computer Engineering — Shanghai, China
  GPA: 3.77/4.00; Outstanding Graduate

Research Interests

I am broadly interested in the intersection of physics and computation, particularly quantum physics, cavity QED, and photonics. I am interested in understanding fundamental physical principles through simulation and modeling, and am especially willing to develop hands-on experimental skills by contributing to the design, construction, and characterization of physical systems. I am motivated not only by understanding how these systems work, but also by building and testing them. I am also interested in developing and applying artificial intelligence and machine learning methods to accelerate physical simulations, analyze complex experimental data, and connect theoretical models with experimental observables.

Research Experience

- **University of Michigan, Department of Radiation Oncology** — May. 2025 - May. 2026
  Undergraduate Research Assistant — Ann Arbor, USA
  - Advisor: Lise Wei
  - Equating the Biologically Effective Dose of SBRT and 90Y SIRT
    * Modeled subvoxel SIRT nonuniformity via EUD, then fit mixed-effects logistic dose–response with covariates; calibrated repair rate and micro-dose CV by aligning SBRT/SIRT curves with bootstrap uncertainty.
    * Co-authored "Toward a Common Biological Dose Scale for SBRT and SIRT Using Imaging-Derived Liver Functional Toxicity Response," presented at the 2026 ASTRO Annual Meeting as a Quick Pitch Oral Scientific Session and published as a conference abstract in the *International Journal of Radiation Oncology, Biology, Physics*.
- **University of Michigan, Deng Research Lab** — Sep. 2024 - Aug 2025
  Undergraduate Research Assistant — Ann Arbor, USA
  - Advisor: Chulwon Lee, Faculty Instructor: Hui Deng
  - Loss–Induced Synthetic Magnetic Field
    * Designed & simulated rotationally asymmetric Si ring–resonator arrays to engineer loss–induced synthetic magnetic fields; observed two-sided non-Hermitian skin-effect localization in 3D FDTD.
    * Built a non-Hermitian tight-binding model; eigen-analysis identified zero net loop phase as the barrier to skin-effect; proposed phase/loss asymmetry and distance-offset remedies and validated designs in simulation.
    * Tools: SYS Lumerical FDTD, MATLAB; GrateLake HPC.
  - Non-Hermitian Exciton-Polariton Transport beyond Lorentz Reciprocity Theorem
    * Fall 2024 - Winter 2025 Undergraduate Research Opportunity Program

* Using Ansys Lumerical to simulate a nanobeam cavity integrated with a grating coupler
* Exfoliating MoSe2 to make monolayer sample
* Manuscripts submitted: Lee, C., Lee, J.H., Zhang, K., Wang, Y., **Wang, Y.**, Lydick, N., Miao, J., Xin, X., Liu, Y., Sun, K., & Deng, H. "Harnessing nanophotonic edgemodes by engineering PT symmetry"

- **Shanghai Jiao Tong University, Laboratory of Ultrafast Integrated Systems (LUIS)** Aug. 2023 - May 2024

Undergraduate Research Assistant Shanghai, China

– Advisor: Xuyang Lu
– Laboratory website: https://sites.gc.sjtu.edu.cn/xuyang-lu/
– ENGR4903J Spring 2024: Chip-scale switch-mode wireless power transfer system based on parity-time method
  * Use the Kuramoto model and Adler equation to design an array of oscillators that can change the mutual phase difference by adjusting the coupling coefficient
  * Understand the basic usage of Cadence and the methods associated with analog circuits
– ENGR4903J Fall 2023: Machine-learning method for mmWave based voice and vital signal recognition
  * Implement the VMD-HHT algorithm to filter noise better and reconstruct the received signal
  * Learn the underlying principles of microcontroller systems, how signals are sent and received between microcontrollers

Projects

- **Coupled Flow Matching (CPFM): Controllable Dimensionality Reduction & Reconstruction** May. 2025 - Oct. 2025

https://arxiv.org/abs/2510.23015 Ann Arbor, USA

– CPFM, a framework that learns coupled continuous flows for data x and low-dim embeddings y to enable both $p(y\,|\,x)$ sampling (latent-space flow) and $p(x\,|\,y)$ reconstruction (data-space flow).
– Extended Gromov–Wasserstein OT with a kernelized quadratic objective to encode semantic priors; derived an alternating minimization with entropic Sinkhorn, achieving $O(n^2)$ per-iteration complexity with an ε-scheduling scheme for stability.

- **Analogical analyses of epidemic evolution from the fluid mechanics view** Sep. 2022 - May. 2024

The 26th 'Shanghai Jiaotong University Student Innovation and Practice Programme' Project Shanghai, China

– Model epidemic evolution using meta cellular automate written in Python and Matlab
– Study second-order phase transitions and link to outbreaks of epidemics

Teaching and Mentoring Experience

– **JI Center for Learning and Teaching, Shanghai Jiao Tong University** Sep. 2023 - Aug. 2024

ENGR151(Accelerated Introduction to Programming), ECE311(Analog Circuits) Teaching Assistant Shanghai, China

* Hold RC to students as ENGR151 Teaching Assistant
* Hold Lab to students as ECE311 Teaching Assistant
* Contribute to the scripts for the teaching team to automate workflows

Technical Skills

Languages: Lumerical, MATLAB, PyTorch, HPC, scientific computing

Miscellaneous: LaTeX, Cadence, Verilog, Russian

Achievements

2026 **James B. Angell Scholar**, University of Michigan
2025 **Michigan Engineering Continuing Student Scholarship**, Roger King Scholarship
2024–2025 **University Honors**, University of Michigan
2024–2025 **Dean's List**, University of Michigan
2024 **SJTU Undergraduate Excellent Scholarship**, Academic Year 2023–2024
2024 **Meritorious Winner**, COMAP Mathematical Contest in Modeling (MCM)
2023 **SJTU Undergraduate Excellent Scholarship**, Academic Year 2022–2023
2020 **First Prize**, Chinese Physics Olympiad (CPhO), Guangdong Province